Problem-solving

Applying secure knowledge in less familiar tasks

Use problem-solving to help students apply secure knowledge when the route is less familiar. Plan the extra decision demand rather than disguise new teaching inside a complex task.

How to do it well

1 Identify the transferable knowledge. Confirm students can use the relevant concepts and procedures in familiar tasks.

2 Select the new demand. Change the context, representation or route choice while keeping the underlying content accessible.

3 Plan and represent. Students identify known information, the goal and a useful diagram or structure.

4 Attempt with purposeful support. Use prompts, comparison or modelling when the search reveals a specific barrier.

5 Compare and generalise. Discuss approaches and the clues that selected them, then apply the reasoning to another task.

What this sounds like

“What do we know, what are we trying to find and which relationship could connect them? Sketch the situation before selecting your approach.”

Classroom example. In Maths, pupils use familiar methods in a task where the approach is not labelled. They represent the problem, identify useful information and justify the selected route. When a barrier concerns a taught relationship, the teacher prompts or models that point. A final comparison explains why alternative routes do or do not work.

Top tips that make the technique work

Check the knowledge base first. Students cannot reason effectively with relationships or methods they have not learned.

Separate selection from execution. Ask which approach fits and why before completing every step.

Teach a useful heuristic explicitly. A diagram, simpler case or organised list can support the task without becoming a universal trick.

Model when search becomes unproductive. Sustained confusion may need guidance rather than more unguided time.

Compare routes after solving. Explain why different approaches work and when each is useful.

Do and avoid

Do	Avoid
Match the unfamiliar demand to secure knowledge.	Calling an untaught procedure problem-solving.
Discuss the reasoning behind the chosen route.	Valuing struggle without attending to whether it produces learning.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

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